

Final exam

1. Determine whether the following series converge or diverge. Show the test you use.

(a) $\sum_{n=1}^{\infty} \frac{3n+1}{n^3}$

(b) $\sum_{n=2}^{\infty} \frac{(-1)^n}{\ln(n)}$

(c) $\sum_{n=1}^{\infty} \frac{5^n}{n!}$

2. Determine whether each series is absolutely convergent, conditionally convergent, or divergent.

(a) $\sum_{n=1}^{\infty} (-1)^{n+1} \frac{n}{n^2+4}$

(b) $\sum_{n=1}^{\infty} \frac{\cos(n)}{n^2}$

3. Find the radius and interval of convergence of the power series.

$$\sum_{n=0}^{\infty} \frac{(x-2)^n}{3^n \sqrt{n+1}}$$

4. Find the first three nonzero terms of the Taylor series of the function about $x = 3$.

$$f(x) = \sqrt{x}$$

5. Write the function as a power series and state its interval of convergence.

$$f(x) = \frac{1}{4-x}$$

Good luck!